

Generated Question Paper

Q1. The advantage of using a pyramid of energy is that it is more efficient. [2 marks]

Q2. (ii) State one recommendation, other than regular brushing, for proper care of teeth. [1 marks]

Q3. (b) (i) Name the source of energy used by organism A. [1 marks]

Q4. (e) shows a synapse and parts of two neurons. (e) begins center end center (i) Draw an arrow to show the direction of the signal across the synapse. [1 marks]

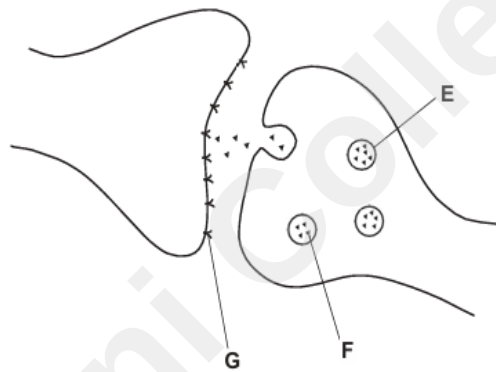


Fig. 4.1

Figure: synapse_diagram.png

Q5. (ii) State the names of the structures labelled A and B on Fig.2. [2 marks]

Q6. (f) Describe the process by which a fertilised egg cell is fertilised. [3 marks]

Q7. (d) Explain why reflexes occur in unconscious people. [1 marks]

Q8. The average photosynthesis rate of soybean plants was measured in two plots with different CO₂ concentrations, 370 ppm and 550 ppm. Describe and explain the effect of carbon dioxide concentration on the average rates of photosynthesis of the soybean plants from 04:00 to 22:00. [6 marks]

The atmospheric carbon dioxide concentrations in the two plots were kept at:

- 370 ppm, which is similar to the current atmospheric carbon dioxide concentration
- 550 ppm, which is a possible future atmospheric carbon dioxide concentration.

When the soybean plants were fully grown, scientists calculated the average rates of photosynthesis at regular intervals from 04:00 to 22:00 for both plots.

The results are shown in Fig. 2.1.

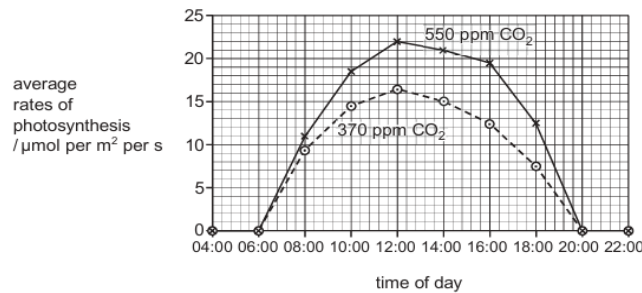


Fig. 2.1

Figure: photosynthesis_rate_graph.png

Q9. (c) Complete Table 2.1 by stating two structural features of epidermis and mesophyll tissues and explain how each feature is an adaptation for photosynthesis. [4 marks]

Q10. (b) State the organs that make up the central nervous system. [1 marks]

Q11. (b) Lugworms live in sand on beaches and feed on diatoms, which require ammonium ions. Researchers tested effects of microplastics on lugworms; see Table 3.1. Describe and explain why researchers concluded that non-biodegradable microplastics are most harmful to lugworms using Table 3.1. [4 marks]

Q12. (d) Name two products of biotechnology, other than bread, that make use of microorganisms. [2 marks]

Q13. (e) Explain the effects of nitrate ion deficiency on plant growth. [3 marks]

Q14. The temperature of the atoms in the nucleus of the atoms is the same as the temperature of the nucleus of the atom. [2 marks]

Q15. (ii) Name the process that causes gas bubbles to form in the dough. [1 marks]

Q16. (iii) Name the gas that forms to create the gas bubbles in the dough. [1 marks]

Q17. The following are the advantages of this method: [3 marks]

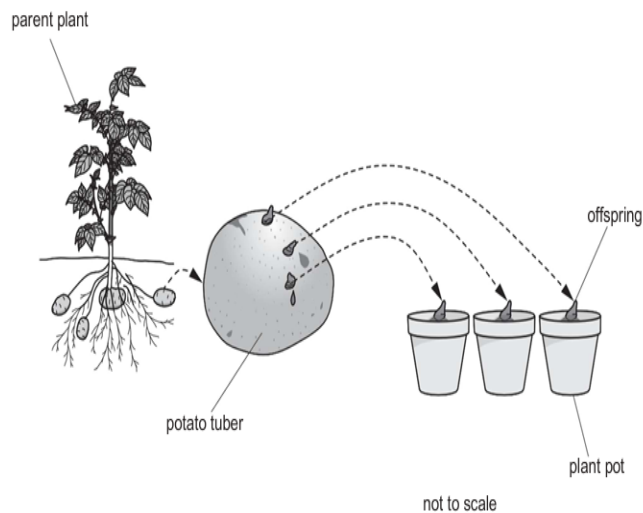


Fig. 6.2

Figure: potato_tubers_method.png

Q18. (ii) Enzymes are the chemical elements found in all proteins. [2 marks]

Q19. (a) Describe how the structure of a neurone relates to its function. [3 marks]

Q20. Enzymes are catalysts. (a) (i) Describe what is meant by the term catalyst. [2 marks]

Q21. (a) State the hormone released from the testes. [1 marks]

Q22. The name of organism A is a comma. [1 marks]

Biotechnology is used in the process of bread-making.

Fig. 1.1 shows some of the steps in making bread.

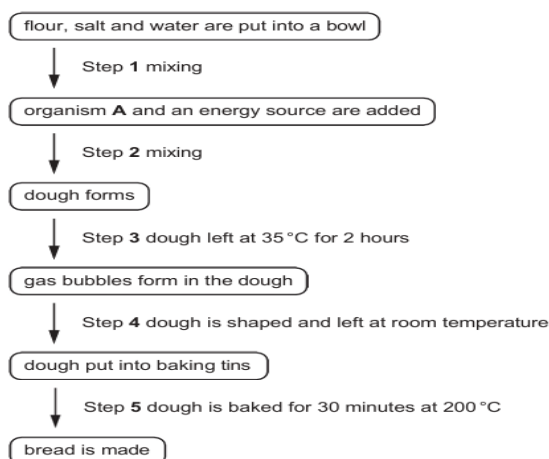


Fig. 1.1

Figure: bread_making_steps.png

Q23. A: Blood vessel C. [2 marks]

Q24. The a) The possible causes of increased atmospheric carbon dioxide. [3 marks]

Q25. Fishing is a way to conserve fish stocks. [2 marks]

Q26. The process of converting ammonium ions into nitrate ions is called the nitrate reaction. [1 marks]

Q27. Scientists had higher breathing rates in the 550 ppm CO₂ plot. [2 marks]

Q28. (f) Discuss the effects of non-biodegradable plastics on terrestrial ecosystems. [5 marks]

Q29. The right ventricle is located at the center of the heart. [1 marks]

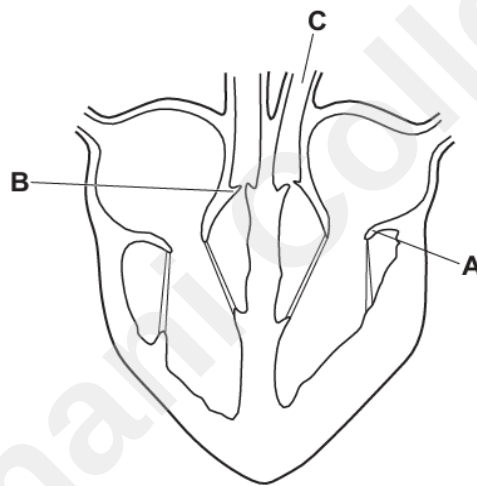


Fig. 2.1

Figure: Fig2_1_HeartCrossSection.png

Q30. A sperm cell is a cell that divides into two individual cells. A sperm cell is a cell that divides into two individual cells. A sperm cell is a cell that divides into two individual cells. [3 marks]



Fig. 5.3

Figure: sperm_cell_micrograph.png

Q31. The oceans contain fish stocks that can be managed as a sustainable resource to provide food for humans. [2 marks]

Q32. The nitrogen cycle is a process that occurs in the nitrogen cycle. [1 marks]

Q33. noindent _____ (ii) State a conclusion for the results shown in Table
 [1 marks]